

GSP313 - Implement Load Balancing on Compute Engine: Challenge Lab

```
# Set environment variables
export REGION=""
export ZONE=""
export PROJECT_ID=""
```

Task 1. Create multiple web server instances

```
# Create three VM instances
gcloud compute instances create web1 \
  --zone=$ZONE \
  --machine-type=e2-small \
  --tags=network-lb-tag \
  --image-family=debian-12 \
  --image-project=debian-cloud \
  --metadata startup-script='#!/bin/bash
apt-get update
apt-get install apache2 -y
service apache2 restart
echo "<h3>Web Server: web1</h3>" | tee /var/www/html/index.html'
```

```
gcloud compute instances create web2 \
  --zone=$ZONE \
  --machine-type=e2-small \
  --tags=network-lb-tag \
  --image-family=debian-12 \
  --image-project=debian-cloud \
  --metadata startup-script='#!/bin/bash
apt-get update
apt-get install apache2 -y
service apache2 restart
echo "<h3>Web Server: web2</h3>" | tee /var/www/html/index.html'
```

```
gcloud compute instances create web3 \
  --zone=$ZONE \
  --machine-type=e2-small \
  --tags=network-lb-tag \
  --image-family=debian-12 \
  --image-project=debian-cloud \
  --metadata startup-script='#!/bin/bash
```

```
apt-get update
apt-get install apache2 -y
service apache2 restart
echo "<h3>Web Server: web3</h3>" | tee /var/www/html/index.html'
```

```
# Create a firewall rule to allow HTTP traffic
gcloud compute firewall-rules create www-firewall-network-lb \
  --allow=tcp:80 \
  --target-tags=network-lb-tag
```

Task 2. Configure the load balancing service

```
# Create a static external IP
gcloud compute addresses create network-lb-ip-1 --region=$REGION
```

```
# Create an HTTP health check
gcloud compute http-health-checks create basic-check
```

```
# Create a target pool
gcloud compute target-pools create www-pool \
  --region=$REGION \
  --http-health-check=basic-check
```

```
# Add instances to the target pool
gcloud compute target-pools add-instances www-pool \
  --instances=web1,web2,web3 \
  --instances-zone=$ZONE \
  --region=$REGION
```

```
# Create a forwarding rule
gcloud compute forwarding-rules create www-rule \
  --region=$REGION \
  --ports=80 \
  --address=network-lb-ip-1 \
  --target-pool=www-pool
```

Task 3. Create an HTTP load balancer

```
# Create an instance template
gcloud compute instance-templates create lb-backend-template \
  --region=$REGION \
  --network=default \
  --subnet=default \
  --tags=allow-health-check \
  --machine-type=e2-medium \
  --image-family=debian-12 \
  --image-project=debian-cloud \
  --metadata startup-script='#!/bin/bash
apt-get update
apt-get install apache2 -y
a2ensite default-ssl
a2enmod ssl
vm_hostname="$(curl -H "Metadata-Flavor:Google" \
<http://169.254.169.254/computeMetadata/v1/instance/name>)"
echo "Page served from: $vm_hostname" | tee /var/www/html/index.html
systemctl restart apache2'

# Create a managed instance group
gcloud compute instance-groups managed create lb-backend-group \
  --template=lb-backend-template \
  --size=2 \
  --zone=$ZONE

# Create a firewall rule to allow health checks
gcloud compute firewall-rules create fw-allow-health-check \
  --network=default \
  --action=allow \
  --direction=ingress \
  --source-ranges=130.211.0.0/22,35.191.0.0/16 \
  --target-tags=allow-health-check \
  --rules=tcp:80

# Create a health check
gcloud compute health-checks create http http-basic-check --port=80

# Set named ports for the instance group
gcloud compute instance-groups managed set-named-ports lb-backend-group \
  --named-ports http:80 \
  --zone=$ZONE
```

Create a backend service

```
gcloud compute backend-services create web-backend-service \
  --protocol=HTTP \
  --port-name=http \
  --health-checks=http-basic-check \
  --global
```

Add backend to the backend service

```
gcloud compute backend-services add-backend web-backend-service \
  --instance-group=lb-backend-group \
  --instance-group-zone=$ZONE \
  --global
```

Create a URL map

```
gcloud compute url-maps create web-map-http \
  --default-service web-backend-service
```

Create a target HTTP proxy

```
gcloud compute target-http-proxies create http-lb-proxy \
  --url-map web-map-http
```

Create an external IP address

```
gcloud compute addresses create lb-ipv4-1 \
  --ip-version=IPV4 \
  --global
```

Create a forwarding rule

```
gcloud compute forwarding-rules create http-content-rule \
  --address=lb-ipv4-1 \
  --global \
  --target-http-proxy=http-lb-proxy \
  --ports=80
```